

Noise Analysis and Improvement of a Permanent Magnet Synchronous Motor by Dynamic Error Budgeting

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Abstract: This paper presents the mechatronic analysis and performance estimation for a single axis rotational servo system with direct-drive permanent magnet synchronous motor (PMSM). To identify the dominant noise sources acting in the motor control system, physical modeling together with dynamic error budgeting are applied. The analysis is carried out for two modes of operation, zero velocity and constant velocity. The current sensor is identified as the major noise source. It is experimentally verified, that an improvement of the current measurement system yields an improvement of the position noise by a factor of 6.3.

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1. INTRODUCTION

Direct-drive permanent magnet synchronous motors (PMSMs) are widely used in servo systems for precision positioning applications, e.g. robots (Thielmann et al., 2010), measuring machines (Hocken and Pereira, 2011) or telescope systems (Gutierrez, 2003; Jian et al., 2012). They eliminate the errors introduced by gear boxes and offer the advantage of high power density and high dynamic performance (Leonhard, 2001; Gieras, 2009). However, to achieve precise motion with errors in the sub- μ rad range, a detailed analysis of the present noise sources and good mechatronic design of the system are necessary.

One of the most precise ways of controlling a PMSM is with vector based control, such as field-oriented control (FOC) (Leonhard, 2001; Merzoug and Naceri, 2008). The compensation of systematic error sources present in such control systems, especially the current control, have attracted quite some attention. Chen et al. (2002) investigates the systematic errors induced by limitations in the motor controller, e.g. CPU word length or finite encoder resolution. To compensate systematic torque ripples from current measurement errors and motor design, a iterative learning control scheme is suggested by Panda et al. (2008). The reduction of torque ripples induced by quantization errors in the current measurement system using dithering and a kalman filter is studied by Zhu and Fujimoto (2014). Kim et al. (2014) investigates a compensation method for offset and scaling errors in the current measurement system of a PMSM by estimation of the voltage error. For precision positioning applications it is important to understand in detail and to minimize all of these systematic disturbances.

However, the effects of stochastic disturbances acting in the drive system have not been analyzed. If not considered in the system design, these disturbances fundamentally limit the achievable precision. The contribution of this paper therefore is, to analyze the stochastic disturbances acting in the FOC system of a PMSM. The analysis is done by dynamic error budgeting (DEB), using power spectral densities (PSDs) of the noise sources, as described by Jabben and van Eijk (2011). As FOC involves several coordinate transformations, special emphasis is put on the treatment of this transformations within the DEB framework.

The remainder of this paper is organized as follows. Section 2 starts with the description of the analyzed system, which is mathematically modeled in Section 3. The derived models are utilized in Section 4 to analyze the influences of the stochastic disturbances by DEB. In Section 5 performance simulations and corresponding experimental validations are presented. Finally, the conclusion is given in Section 6.

2. SYSTEM DESCRIPTION

The system under investigation is a single axis rotational servo system with a direct-drive PMSM. The custom made PMSM has eleven pole pairs and a rated torque of 6.44 Nm. The servo system is used for positioning of an optical imaging system with high precision. To minimize the influence of the servo system on the measurement, a low positioning error is required. For example, telescope systems aim for a root mean square (RMS) position noise of less than 200 nrad. This noise level is one order of magnitude below good astronomical seeing conditions, cf. Schöck et al. (2009). Similar requirements are targeted in other applications of such servo systems.

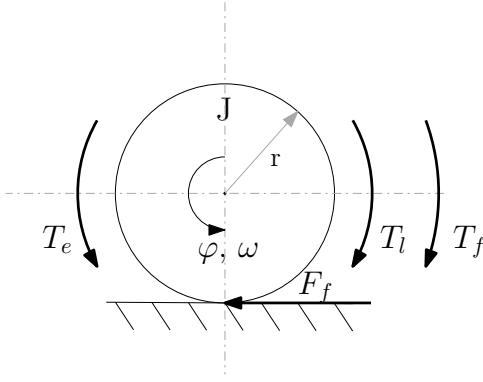


Fig. 1. Lumped mass model of the mechanical system. J is the moment of inertia, T_e the electrical torque, $T_f = F_f r$ the torque due to the friction force F_f and T_l is the load torque.

Generally, two static modes of operation can be defined: i) maintaining a constant position and ii) performing a continuous motion at constant velocity, e.g. for scanning motion. These two modes of operation form the basis for the system modeling and analysis.

3. SYSTEM MODELING

3.1 Physical Model

The lumped mass model shown in Fig. 1 is used to model the mechanical system of one individual positioning axis. The mechanical system is described by

$$\frac{d\varphi}{dt} = \omega \quad (1)$$

$$\frac{d\omega}{dt} = \frac{1}{J} (T_e - T_f - T_l), \quad (2)$$

where φ is the rotor angle, ω the rotor angular velocity, J the moment of inertia, T_e the electrical torque, $T_f = F_f r$ the torque due to the friction force F_f , and T_l is the load torque. As the load is assumed to be well balanced, no constant load torque is present during motion. Therefore T_l can be neglected in the further analysis.

For zero or low velocity, friction is one of the main disturbances. The friction is modeled by the LuGre-Model, according to Åström and Canudas de Wit (2008), given by Equations (3)-(4). This model describes the friction between two bodies by a large number of springs, the so called bristles.

$$\frac{dz}{dt} = \omega - c_0 \frac{|\omega|}{g(\omega)} z \quad (3)$$

$$F_f = c_0 z + c_1 \dot{z} + c_2 \omega, \quad (4)$$

with

$$g(\omega) = F_c + (F_s - F_c) e^{-(\omega/\omega_s)^2}, \quad (5)$$

where z is the average bristle deflection, ω the velocity between the sliding surfaces, F_f the resulting friction force, c_0 represents the stiffness, c_1 the microdamping, c_2 the viscous friction, F_s and F_c the stiction and the Coulomb friction force, and ω_s is a parameter to model the Stribeck effect.

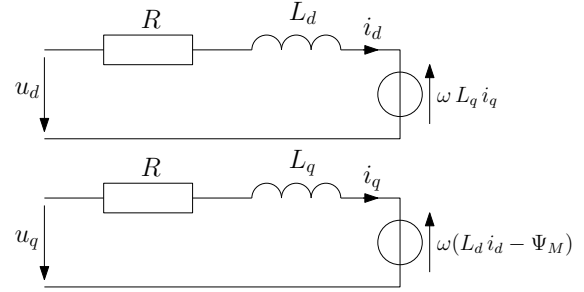


Fig. 2. Schematic representation of the electrical system of a PMSM. By using the Park transform, the system can be described by the direct and quadrature component of the voltage and current.

The combined mechanical model, Equations (1)-(2) and (3)-(4), is linearized for the two static operation modes: i) stick mode with zero velocity and ii) slip mode with constant velocity. In stick mode, the friction force F_f is simplified to

$$F_f = c_0 z + c_1 \dot{z}, \quad (6)$$

resulting in a spring-damper like behavior with a spring constant proportional to c_0 and the damping proportional to c_1 .

In slip mode with constant velocity, F_f is simplified by assuming $\dot{z} = 0$ and $\omega \gg \omega_s$, resulting in a friction force F_f which can be approximated by

$$F_f = F_c \text{sign}(\omega) + c_2 \omega. \quad (7)$$

To describe the electrical part of the PMSM, a model equivalent to Pillay and Krishnan (1988) is used. To simplify the electrical system, all quantities in the three-phase stator coordinate system are transformed by means of the Park transformation (Park, 1929; Lewis, 1958)

$$\mathbf{K}_P = \sqrt{\frac{2}{3}} \begin{bmatrix} \cos(\varphi) & \cos(\varphi - \frac{2\pi}{3}) & \cos(\varphi + \frac{2\pi}{3}) \\ -\sin(\varphi) & -\sin(\varphi - \frac{2\pi}{3}) & -\sin(\varphi + \frac{2\pi}{3}) \end{bmatrix}. \quad (8)$$

Using this transformation, signals in the three-phase stator coordinate system (a, b, c), $\mathbf{x}_{abc}(t) = [x_a(t), x_b(t), x_c(t)]^T$ are transformed to the corresponding signals $\mathbf{x}_{dq}(t) = [x_d(t), x_q(t)]^T$ in the the rotor coordinate system (d, q). Hereby d denotes the direct axis and q the quadrature axis.

The resulting electrical system, as shown in Fig. 2, can be described by

$$\frac{di_d}{dt} = \frac{1}{L_d} (u_d - R i_d + \omega_e L_q i_q) \quad (9)$$

$$\frac{di_q}{dt} = \frac{1}{L_q} (u_q - R i_q + \omega_e L_d i_d - \omega_e \Psi_M), \quad (10)$$

which results in the corresponding electric torque T_e given by

$$T_e = \frac{3}{2} Z_p (\Psi_M i_q + (L_d - L_q) i_d i_q), \quad (11)$$

where L_d, L_q are the d, q axis inductances, u_d, u_q the d, q axis stator voltages, R the stator resistance, i_d, i_q the d, q axis stator currents, Ψ_M the mutual magnetic flux due to the rotor magnets, Z_p the number of pole pairs and $\omega_e = Z_p \omega$ the electrical velocity.

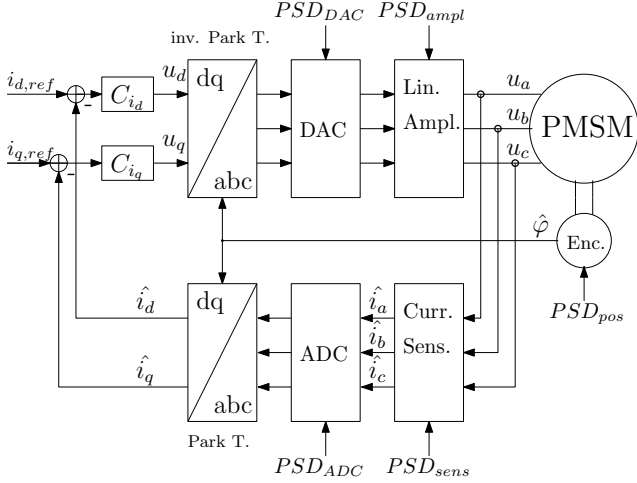


Fig. 3. Schematic representation of the usual current control loop of a PMSM with the individual components and the considered disturbance sources acting on them.

The combined model, Equations (1)-(11), is used to model the physical behavior of one individual axis when performing DEB. The models of the individual components of the remaining control system are derived in the next section.

3.2 Component Models

The inner current control loop of a FOC system is depicted in Fig. 3. The current control system is implemented on a dSpace Rapid Prototyping System (DS1005 and DS5203, dSpace GmbH, Germany). It consists of three 14-Bit digital-to-analog converters (DACs) together with three custom made linear power amplifiers to drive the PMSM. The three phase currents are measured by three Hall effect current sensors and digitized by three 14-Bit analog-to-digital converters (ADCs). The quantities in the three-phase stator coordinate system are then transformed into the rotor coordinate system. This transformation is performed by means of the Park transformation, Equation (8), using the rotor angle, $\hat{\varphi}$, measured by the angular encoder. The current control system is completed by two controllers, one for the direct and one for the quadrature component of the current.

Each of the considered components is modeled by the linear-time invariant (LTI) transfer function and the PSD of the noise induced by the corresponding component. All PSDs are referred to the output of the component.

Digital-to-Analog Converter Model. The transfer function of the 14-Bit, 10Msps DACs, $G_{DAC}(s)$, is modeled by their first order reconstruction filter with a cut-off frequency, $f_{c,DAC}$, of 10 kHz. As the main disturbance, quantization noise is assumed with a PSD given by

$$PSD_{DAC}(s) = |G_{DAC}(s)|^2 \frac{q^2}{6 f_s}, \quad (12)$$

where q is the voltage of the least significant bit (LSB) and f_s the sampling frequency. The quantization noise is treated as white noise with zero mean (Jabben and van Eijk, 2011).

Amplifier Model. The custom made linear power amplifiers use OPA549 operational amplifiers (Texas Instruments Incorporated, Dallas, USA) and are modeled by a first order transfer function, $G_{ampl}(s)$, with a gain, A_{ampl} , of 1.5 and a cut-off frequency, $f_{c,ampl}$, of 110 kHz. To also consider the effects of low frequency disturbances, the noise induced by the amplifiers is modeled as a combination of 1/f-noise and white noise. In accordance with the datasheet, the 1/f-noise is specified by the noise corner frequency, $f_{nc,ampl}$, of 20 Hz and the white, gaussian noise is specified by the calculated noise power spectral density, n_{ampl}

$$PSD_{ampl}(s) = |G_{ampl}(s)|^2 n_{ampl} \left(1 + \frac{(2\pi f_{nc,ampl})^2}{s^2} \right). \quad (13)$$

Current Sensor Model. The current sensors utilize ACS714-05 (Allegro MicroSystems LCC, Worcester, USA) Hall effect current sensors and have a first order transfer function, $G_{sens}(s)$, with a gain, A_{sens} , of 1 V/A and a cut-off frequency, $f_{c,sens}$, of 10 kHz. To include low frequency disturbances, the noise of the current sensors is modeled as a combination of 1/f-noise and white noise. The noise corner frequency, $f_{nc,sens}$, and the noise power spectral density, n_{sens} are determined by measurements (see Table 2). The resulting PSD is given by

$$PSD_{sens}(s) = |G_{sens}(s)|^2 n_{sens} \left(1 + \frac{(2\pi f_{nc,sens})^2}{s^2} \right). \quad (14)$$

Analog-to-Digital Converter Model. The transfer function of the 14-Bit, 10Msps ADCs, $G_{ADC}(s)$, is given by their first order anti-aliasing filter, with a cut-off frequency, $f_{c,ADC}$, of 10 kHz. Quantization noise is assumed to dominate the PSD

$$PSD_{ADC}(s) = |G_{ADC}(s)|^2 \frac{q^2}{6 f_s}. \quad (15)$$

Encoder Model. The transfer function of the encoder, $G_{pos}(s)$, is modeled by a first order low-pass filter, with a cut-off frequency, $f_{c,pos}$, of 5 kHz, and white, gaussian noise, specified by the noise power spectral density, n_{pos} , of 2.93 mrad²/Hz as disturbance

$$PSD_{pos}(s) = |G_{pos}(s)|^2 n_{pos}. \quad (16)$$

Controller Model. As it is common practice, a PI-controller is used for controlling the current in the direct as well as the quadrature axis (Leonhard, 2001; Holmes et al., 2012). The corresponding controller is given by

$$C_{i_q}(s) = K_p + \frac{K_i}{s}. \quad (17)$$

The gains are tuned by simulation to a bandwidth of 650 Hz. This bandwidth is not the maximal achievable bandwidth (Holmes et al., 2012) but rather a compromise between noise amplification and added phase for the superimposed velocity control (see Section 5.2).

4. DYNAMIC ERROR BUDGETING

4.1 Dynamic Error Budgeting

DEB allows to estimate the standstill or constant velocity performance of a system by modeling several stochastic

disturbances acting on the system by their corresponding PSD. The disturbances are then propagated through the LTI models describing the system, cf. Jabben and van Eijk (2011).

The single sided PSD, is used to define the noise power per unit of bandwidth. For a time signal $x(t)$ with zero mean, the PSD is linked with the variance of the time signal by

$$\sigma_x^2 = \int_0^\infty PSD_x(f) df. \quad (18)$$

The PSD at the output y of the system, induced by a specific disturbance x is then given by

$$PSD_{y,x}(s) = |T_{y,x}(s)|^2 PSD_x(s), \quad (19)$$

with the transfer function $T_{y,x}(s)$ between x and y . The overall PSD of the output y can be found by summation of the effect of all disturbances

$$PSD_y(s) = \sum_i |T_{y,x_i}(s)|^2 PSD_{x_i}(s). \quad (20)$$

4.2 System Transformation

To simplify the system, all PSDs and transfer functions in the three-phase stator coordinate system (a, b, c) are transformed to the rotor coordinate system (d, q). Signals, $\mathbf{X}_{abc}(s) = [X_a(s), X_b(s), X_c(s)]^T$, in the stator coordinate system can be transformed to signals, $\mathbf{X}_{dq}(s) = [X_d(s), X_q(s)]^T$, in the rotor coordinate system by the Park transformation, Equation (8),

$$\mathbf{X}_{dq}(s) = \mathbf{K}_P \mathbf{X}_{abc}(s), \quad (21)$$

where $X(s) = \mathcal{L}\{x(t)\}$ denotes the Laplace transform of the corresponding time signal. To perform this transformations two propositions are used.

Proposition 1. Assuming symmetry, i.e. equal transfer functions in all three-phases, then the transfer functions in stator and rotor coordinates are equal and no additional transformation is required.

Proof. The argumentation starts by assuming equal transfer functions in all three-phases,

$$G_{\{a,b,c\}}(s) = G_{abc}(s). \quad (22)$$

Defining a signal in the three-phase stator coordinate system, $\mathbf{Y}_{abc}(s) = [Y_a(s), Y_b(s), Y_c(s)]^T$, for which the individual components are outputs of the LTI transfer functions $G_{abc}(s)$,

$$Y_{\{a,b,c\}}(s) = G_{abc}(s) X_{\{a,b,c\}}(s). \quad (23)$$

and defining a signal in the rotor coordinate system, $\mathbf{Y}_{dq}(s) = [Y_d(s), Y_q(s)]^T$, for which the individual components are outputs of the transfer function $G_{dq}(s)$,

$$Y_{\{d,q\}}(s) = G_{dq}(s) X_{\{d,q\}}(s). \quad (24)$$

The Park transformation, Equation (8), together with Equations (22)-(24) can then be used to obtain

$$\begin{aligned} \mathbf{Y}_{dq}(s) &= \mathbf{K}_P \mathbf{Y}_{abc}(s) = G_{dq}(s) \mathbf{X}_{dq}(s) \\ &= \mathbf{K}_P G_{abc}(s) \mathbf{X}_{abc}(s) = G_{abc}(s) \mathbf{X}_{dq}(s). \end{aligned} \quad (25)$$

From Eq. (25) it can be concluded that the transfer functions in the rotor coordinate system are equivalent to the transfer functions in the stator coordinate system.

$$G_{dq}(s) = G_{\{d,q\}}(s) = G_{\{a,b,c\}}(s) = G_{abc}(s). \quad (26)$$

Proposition 2. Assuming symmetry, i.e. equal PSDs in all three-phases, then the PSDs in stator and rotor coordinates are equal and no additional transformation is required.

Proof. To show the proposition of equal PSDs in stator and rotor coordinates, the disturbances in stator coordinates are assumed to be uncorrelated, gaussian random variables. The variance of a linear combination Y of independent normal random variables X_i (Brase and Brase, 2015)

$$Y = \sum_i a_i X_i + b, \quad (27)$$

is equal to

$$\sigma_Y^2 = \sum_i a_i^2 \sigma_i^2. \quad (28)$$

With the constants, a_i and b , and with σ_i^2 being the variance of X_i and σ_Y^2 being the variance of the linear combination, Y .

By assuming equal PSDs for the disturbances of all three phases,

$$PSD_{\{a,b,c\}}(s) = PSD_{abc}(s), \quad (29)$$

and by using Equation (28), the Park transformation, Equation (8), of the PSDs can be written as

$$\mathbf{PSD}_{dq}(s) = \mathbf{K}_P^{\circ 2} \mathbf{PSD}_{abc}(s), \quad (30)$$

where $\circ 2$ represents the entrywise square and where the PSD in stator coordinates is given by $\mathbf{PSD}_{abc}(s) = [PSD_{abc}(s), PSD_{abc}(s), PSD_{abc}(s)]^T$, together with the PSD in rotor coordinates, given by $\mathbf{PSD}_{dq}(s) = [PSD_{dq}(s), PSD_{dq}(s)]^T$.

Equation (30) can then be reduced to

$$PSD_{dq}(s) = PSD_{abc}(s), \quad (31)$$

by using Equations (32a)-(32b)

$$\cos(\varphi)^2 + \cos(\varphi - \frac{2\pi}{3})^2 + \cos(\varphi + \frac{2\pi}{3})^2 = \frac{3}{2} \quad (32a)$$

$$\sin(\varphi)^2 + \sin(\varphi - \frac{2\pi}{3})^2 + \sin(\varphi + \frac{2\pi}{3})^2 = \frac{3}{2}, \quad (32b)$$

which can be shown by trigonometric equalities.

This means that the PSD in the stator coordinate system is equal to the PSD in the rotor coordinate system. Therefore no additional transformation of the transfer functions or the PSDs are required.

4.3 Transfer Functions

The schematic representation of the transformed current control system for the torque producing quadrature current is shown in Fig. 4. Also shown are the considered disturbances stemming from the DACs, the analog amplifiers, the encoder, the current sensors and the ADCs. As the noise induced by the Park transformation itself can be neglected for this system (cf. Riel et al. (2016)), it is not considered in this analysis. Based on this representation and with the simplified notation $G(s) = G$, the transfer functions of the disturbances to the rotor position, φ , are given by

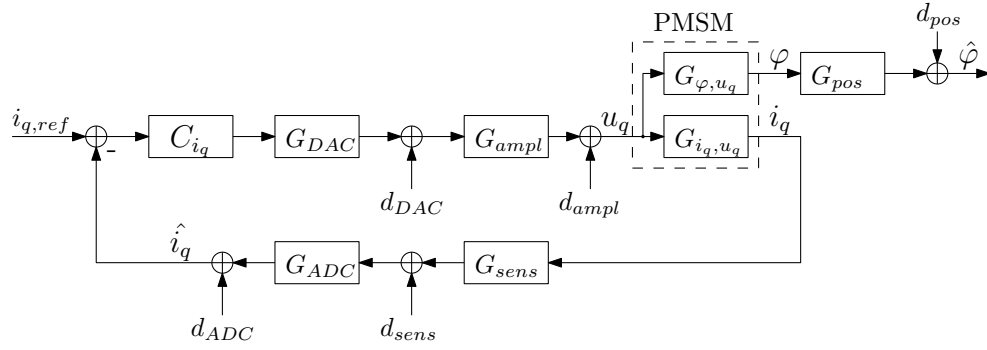


Fig. 4. Schematic representation of the current control loop in rotor coordinates for the torque producing I_q -loop with the individual components and considered disturbance sources. The I_d -loop is equal to the I_q -loop.

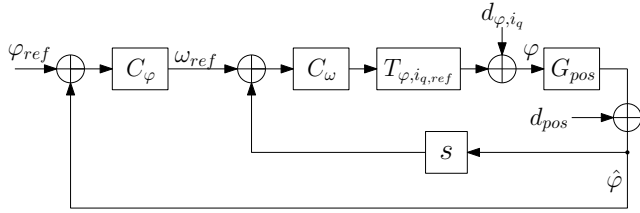


Fig. 5. Schematic of the cascaded FOC system with the transfer function of the current control, $T_{\varphi, i_{q,ref}}$.

$$T_{\varphi, ampl} = \frac{G_{\varphi, u_q}}{1 + C_{i_q} G_{DAC} G_{ampl} G_{i_q, u_q} G_{sens} G_{ADC}} \quad (33)$$

$$T_{\varphi, DAC} = G_{ampl} T_{\varphi, ampl} \quad (34)$$

$$T_{\varphi, ADC} = -C_{i_q} T_{\varphi, DAC} \quad (35)$$

$$T_{\varphi, sens} = G_{ADC} T_{\varphi, ADC}, \quad (36)$$

where G_{i_q, u_q} and G_{φ, u_q} represent the PMSM model.

Using the models of Section 3 and the transfer functions, Equations (33)-(36), the effect of the disturbances acting in the current control system on the rotor position can be calculated. In the next step, the cascaded control system is extended by a velocity and position controller.

4.4 Velocity and Position Control

To analyze the influence of velocity and position control, the current control system is replaced by its closed loop transfer function

$$T_{\varphi, i_{q, ref}} = \frac{C_{i_q} G_{DAC} G_{ampl} G_{\varphi, u_q}}{1 + C_{i_q} G_{DAC} G_{ampl} G_{i_q, u_q} G_{sens} G_{ADC}}. \quad (37)$$

The disturbances acting in the I_q -loop are all transferred to the output and replaced by their combination, d_{φ, i_q} , with the corresponding PSD_{φ, i_q} . The effect of the disturbances in the I_d -loop are neglected, because they produce essentially no torque.

To complete the cascaded control system, the system is then expanded by a velocity and position controller, as shown in Fig. 5. The transfer function of the disturbance acting on the encoder to the rotor angle, φ , is

$$T_{\varphi, pos} = \frac{1}{1 + (s + C_\varphi) G_{pos} C_\omega T_{i_{q, ref}, \varphi}}. \quad (38)$$

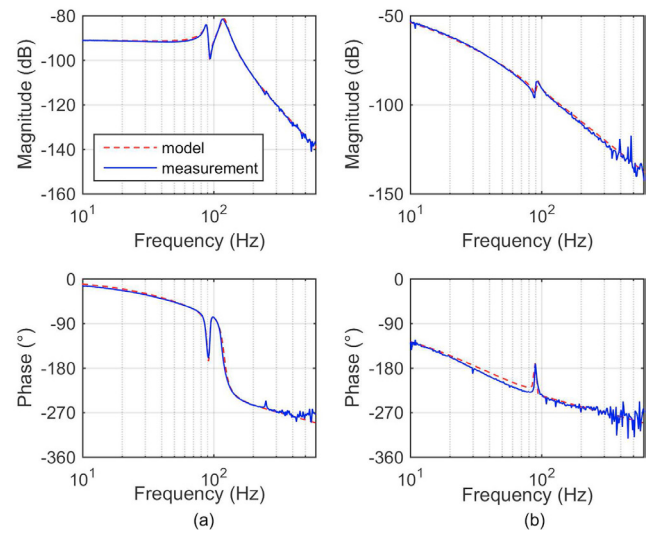


Fig. 6. Bode diagram of the physical model and the measured frequency response data from the quadrature input voltage, u_q , to the rotor angle, φ , for (a) zero velocity and for (b) constant velocity.

The sensitivity function of the position control system on the disturbance from the current control system is given by

$$S_{\varphi, i_q} = -\frac{(s + C_\varphi) C_\omega T_{\varphi, i_{q, ref}}}{1 + (s + C_\varphi) G_{pos} C_\omega T_{\varphi, i_{q, ref}}}. \quad (39)$$

Velocity and position control will attenuate disturbances from the current control system up to their bandwidth.

5. DYNAMIC ERROR BUDGET OF THE PMSM

In the next step the derived models are parametrized and the performance of the servo system is simulated and validated by measurements.

5.1 Characterization

The parameters of the used servo system are estimated using the physical model of Section 3 and least squares fitting using MATLAB Optimization Toolbox. The resulting parameters are given in Table 1, while the known parameters of the remaining components are given in Table 2. A comparison between the measured frequency response data and the model are shown in Fig. 6. In

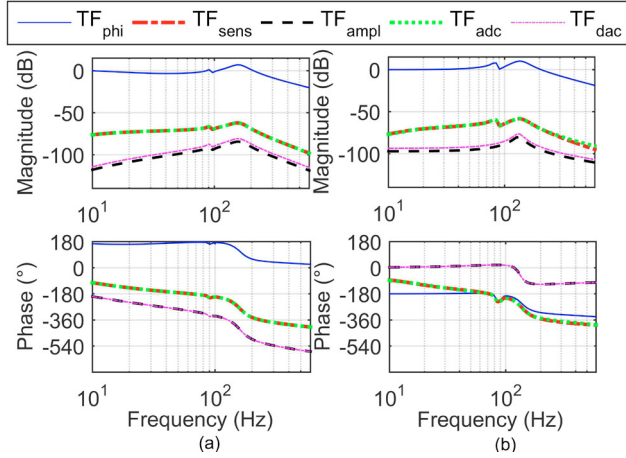


Fig. 7. Bode diagram of the different transfer functions from disturbance sources to the rotor angle of the PMSM for (a) zero velocity and (b) constant velocity.

addition to the behavior predicted by the physical modeling, a resonance/anti-resonance at 90Hz is fitted, which is modeling a structural mode of the system. The derived physical model is in good agreement with the measured system response for both modes of operation. It is also visible that the system shows a very different behavior in the two modes of operation, justifying the additional modeling effort. Using this parametrization of the model, the transfer functions of all considered disturbance sources can be evaluated.

Table 1. Estimated system parameters used for simulation

Number of pole pairs Z_p	11
Resistance R	1.31 Ω
Inductance L_q	3.3 mH
Torque constant K_m	1.81 N · m/A
Inertia J	0.009 kg · m ²
Stiffness c_0	2.4 · 10 ³ N · m/rad
Damping c_1	0.33 N · m/(rad/s)
Viscosity c_2	0.13 N · m/(rad/s)
Coulomb friction F_c	0.01 N · m

Table 2. System parameters of the amplifier and current sensor used for simulation

Bandwidth ADC $f_{c,ADC}$	20 kHz
Bandwidth DAC $f_{c,DAC}$	20 kHz
Sampling frequency f_s	100 kHz
LSB voltage q	610 μ V
Bandwidth amplifier $f_{c,ampl}$	110 kHz
Noise corner amplifier $f_{nc,ampl}$	20 Hz
Amplifier noise PSD n_{ampl}	107 nV ² /Hz
Amplifier gain A_{ampl}	1.5
Bandwidth current sensor $f_{c,sens}$	20 kHz
Noise corner sensor $f_{nc,sens}$	10 Hz
Current sensor noise PSD n_{sens}	10.8 · 10 ⁻⁹ A ² /Hz
Current sensor gain A_{sens}	1 V/A
Bandwidth encoder $f_{c,pos}$	5 kHz
Encoder noise PSD n_{pos}	2.93 nrad ² /Hz
Angular velocity ω	3.6 mrad/s
Current control bandwidth BW_i	650 Hz
Velocity control bandwidth BW_ω	220 Hz
Position control bandwidth BW_φ	45 Hz

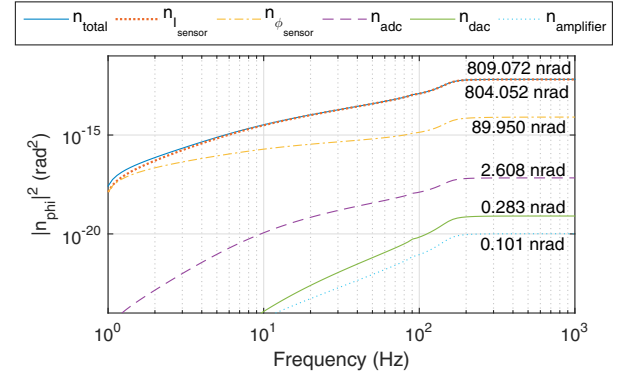


Fig. 8. Cumulative power spectral density for zero velocity.

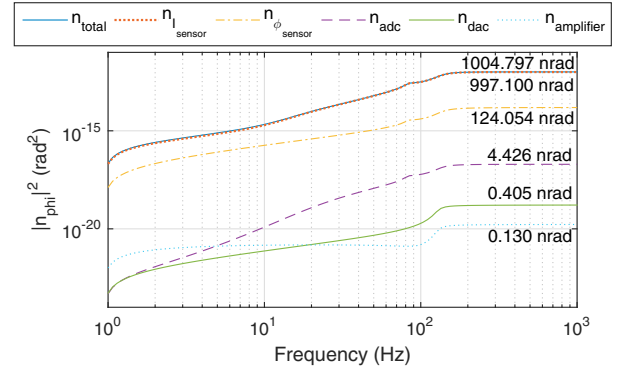


Fig. 9. Cumulative power spectral density for constant velocity.

5.2 Simulation

The simulation is performed with a current control bandwidth of 650 Hz, a velocity control bandwidth of 220 Hz and a position control bandwidth of 45 Hz. The resulting transfer functions are shown in Fig. 7 for zero velocity and a constant velocity of 3.6 mrad/s respectively. The chosen velocity arises from a requirement of the application targeted by the funding project. These transfer functions are used to calculate the PSD of each disturbance source and then the combined PSD of all disturbances. For better visibility of the contributing error sources, the PSDs are plotted as cumulative power spectral densities (CPSDs) in Fig. 8 and Fig. 9 for zero velocity and constant velocity respectively.

In case of zero velocity, Fig. 8 shows that the dominant disturbance source is the noise of the current sensor, with the encoder noise accounting for only 11% of the total position noise. It is also visible that, in this operation regime the position error specification of 200 nrad is already exceeded.

The case of constant velocity of 3.6 mrad/s is depicted in Fig. 9, showing an increased noise level. This difference can be explained by the higher friction force in stick mode, which is opposing all disturbances from the control system. The remaining error is close to 1 μ rad and is dominated by the disturbance due to the current sensor.

To reach the desired position accuracy, further reduction of the position noise is necessary. This can be achieved either by an increased control bandwidth or by a reduction of the present noise sources. Using PI control, no further increase

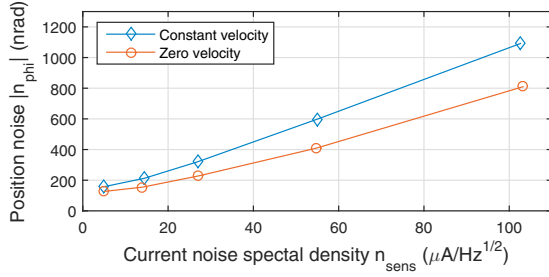


Fig. 10. Simulated relationship between the noise spectral density of the current sensor and the position noise of the rotor for constant and zero velocity.

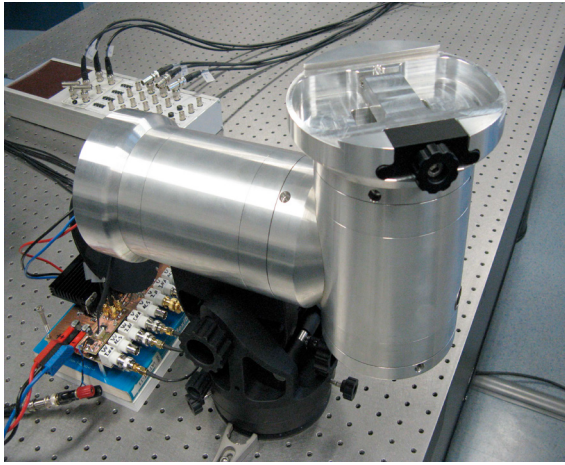


Fig. 11. Laboratory prototype of a single axis rotational servo system with direct-drive PMSMs.

of the control bandwidth is reached. As a consequence, a reduction of the current sensor noise is considered. The resulting effect is simulated using the developed model and depicted in Fig. 10. This result is used to guide the design process to find a suitable current sensor. To test this simulation result, an improved current sensor with reduced noise is implemented. The experimental validation is given in the next section.

5.3 Experimental Validation

To validate the presented simulation results, an experimental setup is used. The experimental setup consists of a prototypical servo system, as shown in Fig. 11 and described in Section 3. Two different types of current sensors are implemented and the resulting performance is compared. The original current sensor is based on the Hall effect and has a noise spectral density of $104 \mu\text{A}/\sqrt{\text{Hz}}$. The improved current sensor, INA225 (Texas Instruments Incorporated, Dallas, USA), uses a shunt resistor with $5 \text{ m}\Omega$ to measure the phase current and has a noise spectral density of $13 \mu\text{A}/\sqrt{\text{Hz}}$. The PI controllers are tuned to the simulated control bandwidth.

Fig. 12 shows a comparison of the position error for the two different current sensors at zero velocity. By using the improved current sensor, the standard deviation of the position error is reduced from 875 nrad to 218 nrad. Correcting for the additional encoder noise present in the measurement of the rotor angle, $\hat{\phi}$, by subtracting the

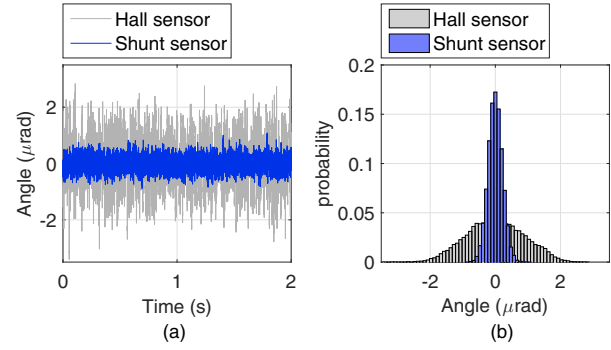


Fig. 12. The measured position error, $\hat{\phi}$, for two different current sensors and zero velocity. Shown are (a) Time signal and (b) histogram of the measured data.

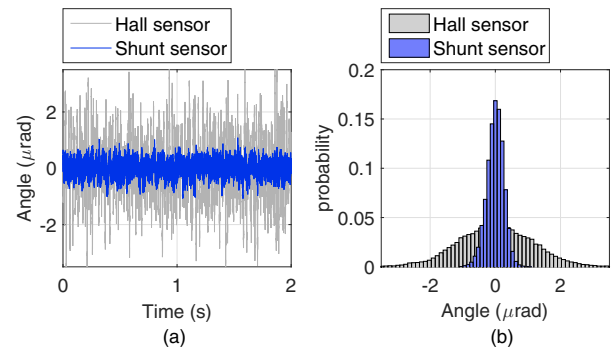


Fig. 13. The measured position error, $\hat{\phi}$, for two different current sensors and a constant velocity of 3.6 mrad/s. Shown are (a) Time signal and (b) histogram of the measured data.

corresponding noise power added by the encoder, the true position error is reduced to 135 nrad. This is a reduction by a factor of 6.3 as compared to the original current sensor and meets the targeted specification.

Fig. 13 shows the comparison between the two sensors for a constant velocity of 3.6 mrad/s. For constant velocity the standard deviation of the position error is reduced from 1124 nrad to 256 nrad. Correcting for the additional encoder noise, the position error is reduced by a factor of 5.8 to 190 nrad. These results also suggest that the simulation consistently underestimates the position noise. This may be due to an unconsidered disturbance source or unconsidered correlation between the noise sources. The reduction of the measurement noise from the current sensors results in an improvement of the system performance by a factor of up to 6.3.

6. CONCLUSION

This paper shows the noise analysis and improvement of a single axis rotational servo system with direct-drive PMSM. It is demonstrated that DEB can be applied to a system which involves several coordinate transformations. Physical modeling and DEB are used to identify the main disturbance source in the FOC system of a PMSM. It is found that the current measurement system is the dominant noise source in the system. The improvement of the used current sensors is proposed and the necessary

improvement calculated. An experimental comparison between two current sensor types with different noise power is carried out. The experimental validation shows good agreement with the simulated results. An improvement of the RMS position noise by a factor of up to 6.3 is achieved.

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